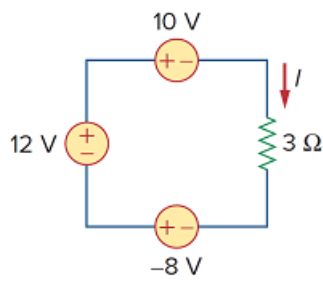


Problem

From the circuit in Fig. 2.83, find I , the power dissipated by the resistor, and the power supplied by each source.

Figure 2.83:



Step-by-step solution

Step 1 of 5

Refer to Figure 2.83 in the textbook.

Apply Kirchhoff's voltage law to the circuit to calculate the current, I .

$$-12 + 10 + 3I - (-8) = 0$$

$$6 + 3I = 0$$

$$I = -2 \text{ A}$$

Thus, the value of the current, I is $\boxed{-2 \text{ A}}$.

Step 2 of 5

Calculate the power dissipated by the resistor, $p_{3\Omega}$.

$$p_{3\Omega} = I^2 \times 3$$

Substitute -2 A for I .

$$p_{3\Omega} = (-2)^2 \times 3$$

$$= 4 \times 3$$

$$= 12 \text{ W}$$

Thus, the power dissipated by the resistor is $\boxed{12 \text{ W}}$.

Step 3 of 5

The power supplied is multiplication of voltage and current when the current enters through negative voltage terminal.

Calculate the power supplied by the 12 V voltage source, $p_{12 \text{ V}}$.

$$p_{12 \text{ V}} = V_s I$$

$$= (12)(-2)$$

$$= -24 \text{ W}$$

Thus, the power supplied by the 12 V voltage source is $\boxed{-24 \text{ W}}$.

Step 4 of 5

Calculate the power supplied by the 8V voltage source, $p_{8\text{V}}$.

$$\begin{aligned} p_{8\text{V}} &= V_s I \\ &= (-8)(-2) \\ &= 16\text{ W} \end{aligned}$$

Thus, the power supplied by the 8V voltage source is $\boxed{16\text{ W}}$.

Step 5 of 5

Calculate the power supplied by the 10V voltage source, $p_{10\text{V}}$.

$$\begin{aligned} p_{10\text{V}} &= V_s I \\ &= -(10)(-2) \\ &= 20\text{ W} \end{aligned}$$

Thus, the power supplied by the 10V voltage source is $\boxed{20\text{ W}}$.